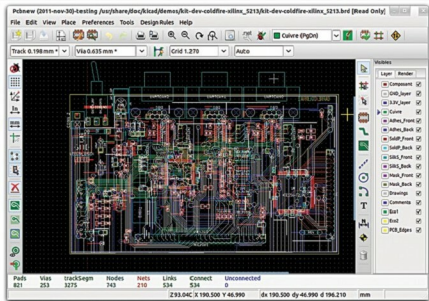


EDA Tools For Circuit Design

Using EDA tools to design is one of the best ways to ensure reliable circuits or systems in later stages of the design cycle. But are you confused by the numerous EDA software tools that you get for free in EFY Plus DVD? This article might help you to select the best ones for your use



KiCad EDA software suite

ANAGHA P.

Creation of a device or instrument involves many stages, right from inception of the idea to getting the final product in hand. Many of these devices have been made simpler by using the free or licensed electronics design and automation (EDA) tools available. Let us take a look at the different processes involved in an electronics design and the software used to automate these processes.

Normally, the steps that need to be taken for designing an electronic product are:

1. Capturing of product idea as schematic
2. Synchronising with printed circuit board (PCB) layout tool
3. Performing the layout (placement and routing)

- #### 4. Creating the manufacturing outputs

Circuit design and schematic entry

The process of designing standard circuit diagrams with related symbols and parameters is known as schematic capture or schematic entry. This is achieved with the help of an EDA software known as schematic editor. Present-day schematic editors also have some or all of the following features:

1. Extensive component libraries
2. Options to create and define your own components
3. Capabilities for hierarchical designing
4. Automatic detection of unfeasible connections and other errors
5. Documentation of circuit details and calculation of bill of materials

6. Creation of netlists and other common formats for storing and transferring data related to the circuits

Creating a layout and printing it on board used to be a fully manual procedure but, thanks to the numerous PCB layout tools available these days, most of the steps in PCB layout creation and editing are automatic now.

For instance, the tools for signal integrity (SI) measure a set of parameters to check the quality of a PCB. Issues like transmission losses, crosstalk, impedance mismatch, thermal stress and strain etc are identified and steps taken to solve these before sending the product for manufacture. This reduces the failure rate of PCBs, thereby reducing time, effort, material and cost.

There are several design rules for minimum spacing between pads and traces, acute angle, minimum annular ring, drill-to-drill clearance etc, which determine whether a product can actually be manufactured properly. The computer-aided design (CAD) and computer-aided manufacturing (CAM) validation ensures that the design created is compatible with all these criteria and lets you create an optimised model. A netlist is generated from this software and sent to a manufacturer for PCB validation.

The right tool for you

So now you have an idea of the different steps of software-enabled circuit designing. The next question is how to select the right tool for your application? What all parameters are to be considered before going to the vendors?